HW5 Math 638 Differential Geometry II, Winter 2026

Due on Wednesday, February 11, 2026 on class.

Q5.1. ([Lee, 2nd, 10-12, p. 270]) Let $\pi : E \to M^n$ and $\tilde{\pi} : \tilde{E} \to M$ be two smooth rank-k vector bundles over a smooth manifold M without boundary. Suppose $\{U_\alpha\}_{\alpha\in A}$ is an open cover of M such that both E and $\tilde{E}$ admit smooth local trivializations over each U_α. Let $\{\tau_{\alpha\beta}\}$ and $\{\tilde{\tau}_{\alpha\beta}\}$ denote the transition functions determined by the given local trivializations of E and $\tilde{E}$, respectively. Show that E and $\tilde{E}$ are smoothly isomorphic over M if and only if for each $\alpha \in A$ there exists a smooth map $\sigma_\alpha : U_\alpha \to GL(k, \mathbb{R})$ such that

$$\tilde{\tau}_{\alpha\beta}(p) = \sigma_\alpha(p) \cdot \tau_{\alpha\beta}(p) \cdot \sigma_\beta(p)^{-1} \qquad \text{for each } p \in U_\alpha \cap U_\beta.$$

Q5.2. ([Lee, 2nd, 10-6 modified, p. 269]) (Vector bundle construction theorem): Let M^n be a smooth manifold without boundary, and let $\{U_\alpha\}_{\alpha\in A}$ be an open cover of M. Suppose for each $\alpha, \beta \in A$ we are given a smooth map $\tau_{\alpha\beta}$: $U_\alpha \cap U_\beta \to GL(k; \mathbb{R})$ such that

$$\tau_{\alpha\beta}(p) \cdot \tau_{\beta\gamma}(p) = \tau_{\alpha\gamma}(p) \qquad \text{for each } p \in U_\alpha \cap U_\beta \cap U_\gamma$$

is satisfied for all $\alpha, \beta, \gamma \in A$. Show that there is a smooth rank-k vector bundle $\pi : E \to M$ with smooth local trivializations $\Phi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times \mathbb{R}^k$ whose transition functions are the given maps $\tau_{\alpha\beta}$.
(Hint: Define an appropriate equivalence relation on $\sqcup_{\alpha\in A}(U_\alpha \times \mathbb{R}^k)$, and use the vector bundle chart lemma.]

Q5.3. ([Lee, 2nd, 10-9 modified, p. 270]) (Extension lemma for sections of restricted bundles): Suppose M^n is a smooth manifold, $\pi : E \to M$ is a smooth vector bundle of positive rank, and $S \to M$ is an embedded submanifold without boundary.

3a) For any smooth section σ of the restricted bundle $E|_S \to S$, show that there exist a neighborhood U of S in M and a smooth section $\tilde{\sigma}$ of $E|_U$ such that $\sigma = \tilde{\sigma}|_S$.

3b) Show that every smooth section of $E|_S$ extends smoothly to all of M if and only if S is properly embedded. (Hint: Compare to [Lee, 2nd, Lemma 2.26, p. 45]).

Q5.4. ([Lee, 2nd, 10-15, p.270) Let V be a finite-dimensional real vector space, and let $G_k(V)$ be the Grassmannian of k-dimensional subspaces of V (see [Lee, 2nd, Example 1.36]). Let T be the subset of $G_k(V) \times V$ defined by

$$T = \{(S, v) \in G_k(V) \times V,\ v \in S\}.$$

Show that T is a smooth rank-k subbundle (called tautological vector bundle) of the product bundle $G_k(V) \times V \to G_k(V)$, and is thus a smooth rank-k vector bundle over $G_k(V)$.